

"ROUGH SHEET"

Lob-01

Course No : Physics 1213

Course Title: physics

Experiment No : 01

Experiment name : Determination of the specific rotation of sugar solution by means of polarimeter

Date: 23.11.2025

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Section - C

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Incomplete
calculations

~~23.11.25~~

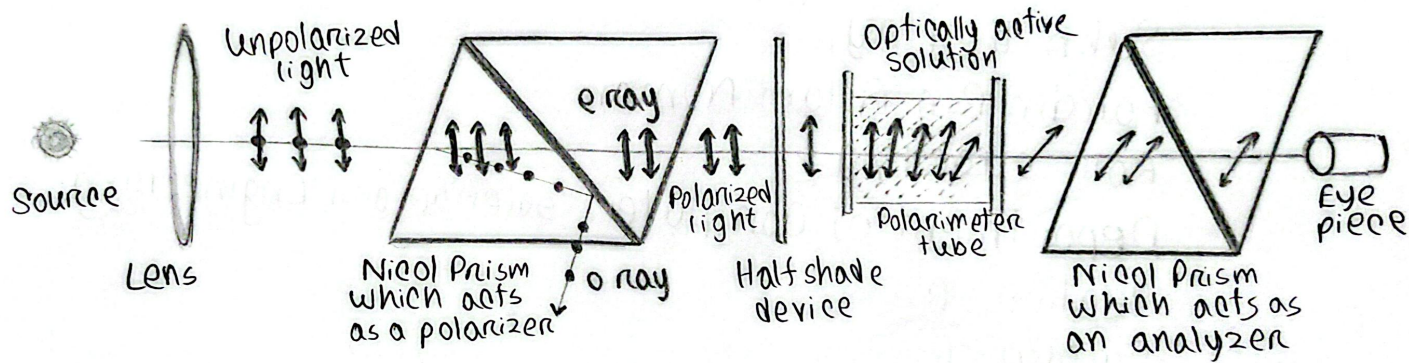


Fig. 1: Laurent's half shade polarimeter

Experiment No: 01

Experiment Name: Determination of the specific rotation of sugar solution by means of polarimeter

Theory:

The specific rotation of a solution is a characteristic physical property of a given optically active molecule. If the specific rotation of a compound is positive, the compound is said to be dextrorotatory, if negative, it is referred to as levorotatory. The substances that rotate the plane of vibration to the right called right-handed or dextrorotatory and the substances that rotate the plane of vibration to the left are called left-handed or levorotatory. The angle of rotation produced to the plane of vibration by an optically active substance in solution is proportional to,

1. The thickness of solution
2. The concentration of the solution or density of active substance in the solvent
3. The nature of the substance

Thus we can write,

$$\theta \propto lc$$

$$\Rightarrow \theta = Slc$$

$$\Rightarrow S = \frac{\theta}{lc}$$

If l is in cm, then,

$$S = \frac{10\theta}{lc}$$

Here,

θ = Angle of rotation

l = Length of solution in decimeters

$c = \frac{m}{V}$ = concentration of solution ($\frac{g}{cm^3}$)

m = mass of sugar (g) dissolved in water

V = Volume of sugar solution

S = Specific rotation

$$\left[\frac{20 \text{ div}}{19^\circ} \right]$$

determ.
15

calculations:

Vernier constant, $V_c = \text{MSD} - \text{VSD}$

$$= \frac{20}{20} \text{ VSA}$$

$$= \frac{20 - 19}{20}$$

$$= \frac{1}{20} = 0.05 \text{ MSD}$$

$$\left(\frac{19}{20} \right)^\circ$$

$$VSR = V_c \times VSD = 0.05 \times 14 \text{ div}$$

$$MSR = 15 \text{ deg}$$

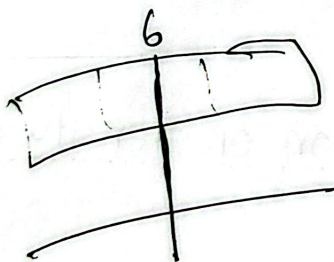
$$I.S.D. \text{ of M.S.} = 1^\circ$$

$$VSR = V_c \times d.v. = \text{degree}$$

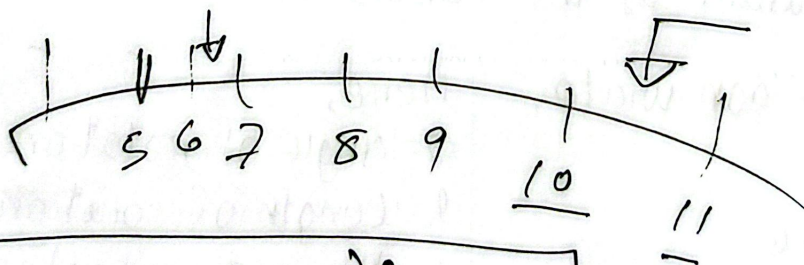
$$I.S.D. \text{ of V.S.} = \left(\frac{19}{20} \right)^\circ$$

$$V.C. = \left(\frac{19}{20} \right)^\circ (1^\circ) - \left(\frac{19}{20} \right)^\circ$$

$$= \left(\frac{1}{20} \right)^\circ$$



$$S = -$$



$$M.S.R = ()^\circ$$

$$V.S.R = \text{div}$$

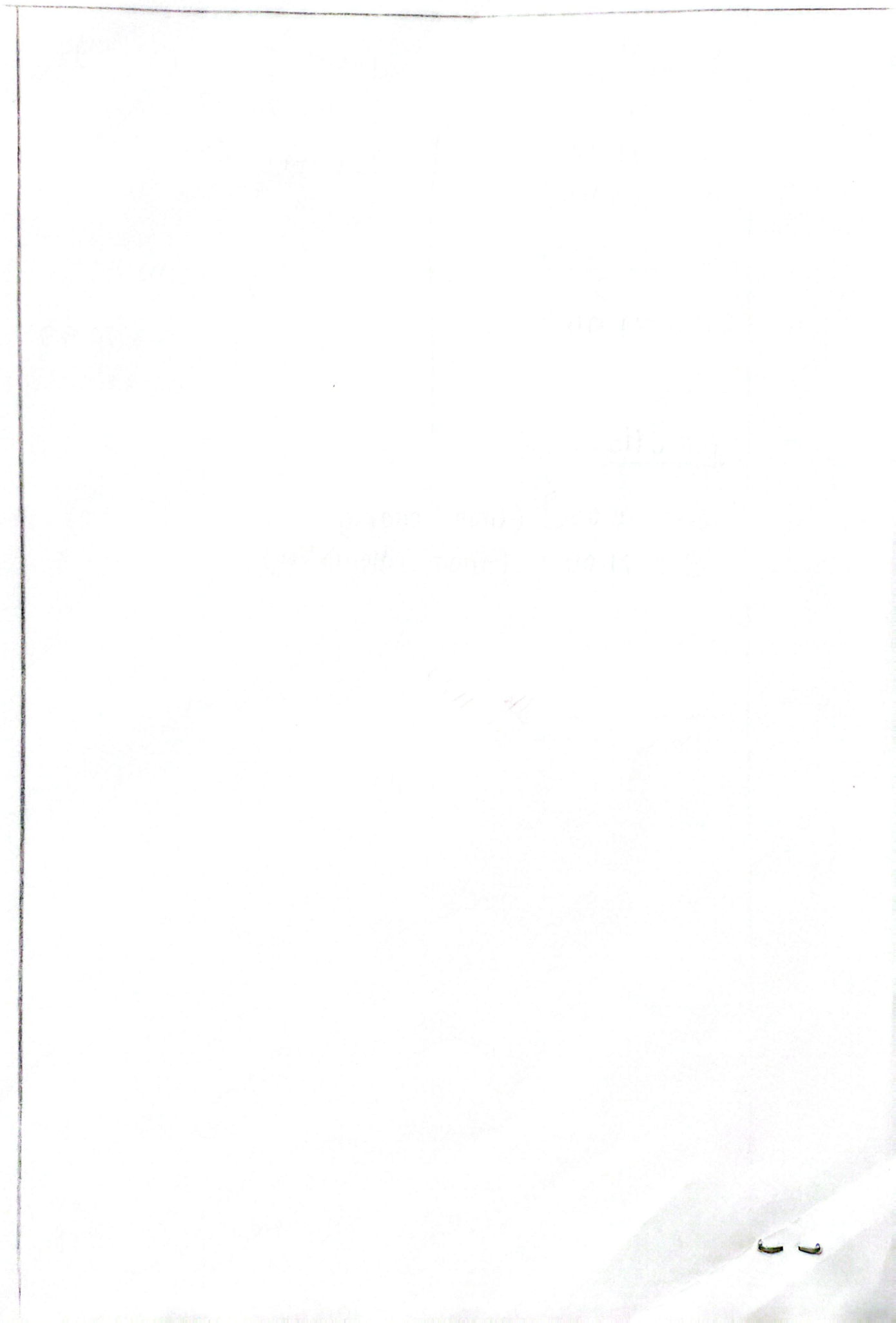
$$TR =$$

$$M.S.R \pm 0^\circ$$

$$M.S.R = 6^\circ$$

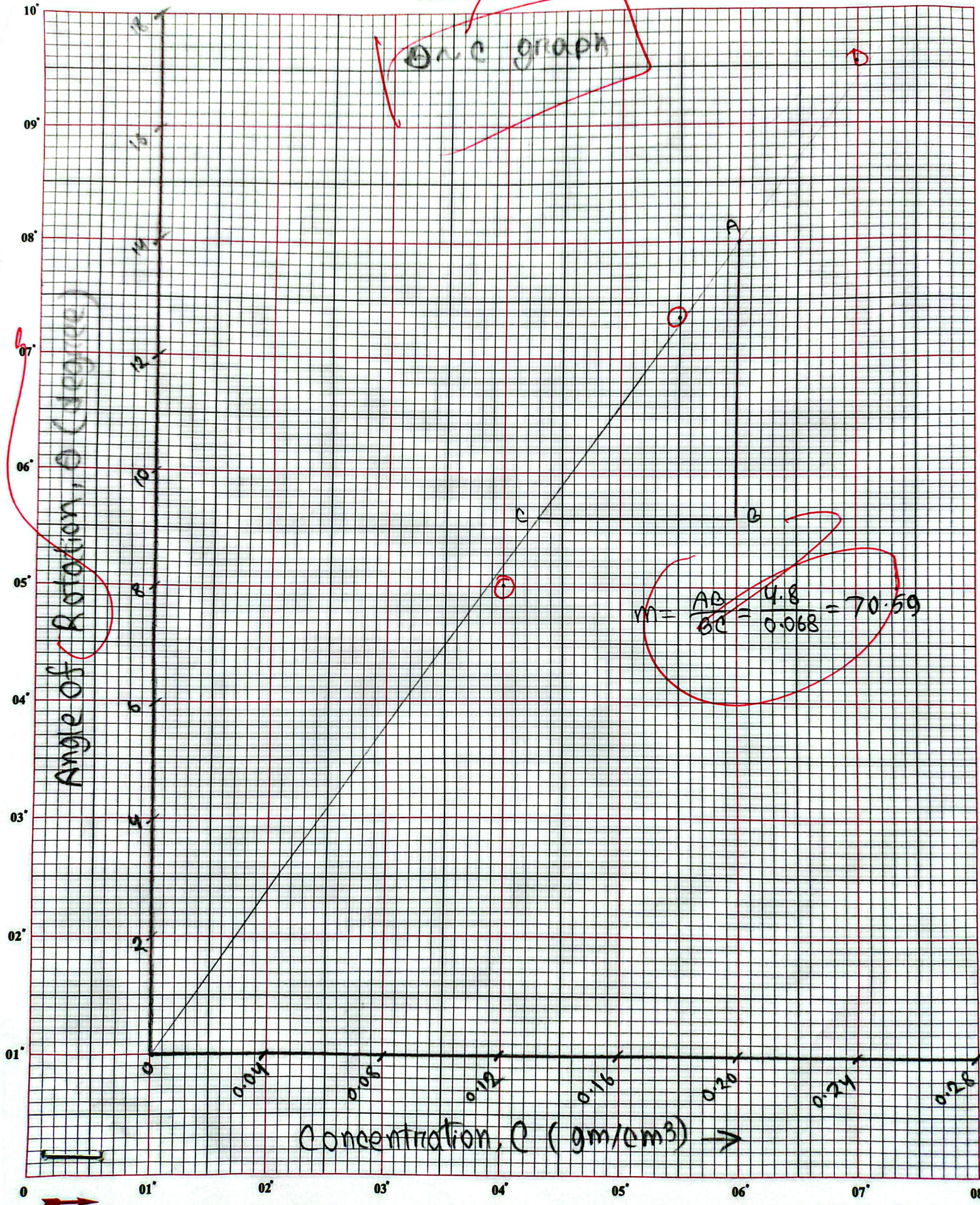
$$V.S.R = 6 \text{ div}$$

$$= 6 \times V.C.$$



Name: Roll No:

$\theta \sim c$ graph



Calculations:

$$S_1 = 72.7^\circ$$

$$S_2 = 71.99^\circ$$

$$S_3 = 68.45^\circ$$

$$S = \frac{S_1 + S_2 + S_3}{3}$$
$$= 71.05^\circ$$

From graph plotting,

$$m = 70.59$$

We know,

$$S'' = \frac{10\theta}{lc}$$

$$= \frac{10}{l} \times m$$

$$= \frac{10}{9.8} \times 70.59$$
$$= 72.03^\circ$$

Results:

$$S = 72.03^\circ \text{ (from graph)}$$

$$S = 71.05^\circ \text{ (from calculation)}$$

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Required Apparatus:

- | | |
|-----------------------|--------------------|
| 1. Polarimeter | 5. Beaker |
| 2. Balance | 6. Thermometer |
| 3. Measuring cylinder | 7. Source of light |
| 4. Sugar | |

Experimental Data:

Table 1 - Data for determining angular rotation

No. of Obs	Strength of solution (%)	Concentration of sugar solution (gm/cm ³)	Reading with water p (deg)				Reading with solution Q (deg)				Angular Rotation (Q-P) (deg)	Mean Angular Rotation (deg)	Specific Rotation (deg)
			MSR (deg)	VSD (div)	VSR (deg)	TR (deg)	MSR (deg)	VSD (div)	VSR (deg)	TR (deg)			
1	21	0.24	Scale A	3	10	0.5	3.5	Scale A	20	12	0.6	20.6	17.1
			Scale B	3	10	0.5	3.5	Scale B	20	12	0.6	20.6	17.1
2	18	0.18	Scale A	3	10	0.5	3.5	Scale A	16	4	0.2	16.2	12.7
			Scale B	3	10	0.5	3.5	Scale B	16	4	0.2	16.2	12.7
3	12	0.12	Scale A	3	10	0.5	3.5	Scale A	11	10	0.55	11.55	8.05
			Scale B	3	10	0.5	3.5	Scale B	11	10	0.55	11.55	8.05
4			Scale A	3	10	0.5	3.5	Scale A					
			Scale B	3	10	0.5	3.5	Scale B					

~~MSR - Main Scale Reading~~

~~VSD = Vernier Scale Division~~

~~VSR = Vernier Scale Reading~~

~~TR = Total Reading~~

Discussion: In this experiment, the specific rotation of sugar solution (s) was determined using a polarimeter. By measuring the angle of rotation produced when polarized light passed through the solution. The specific rotation was calculated using the standard formula,

$$s = \frac{100\theta}{lc}$$

Precautions:

- The polarimeter tube was well cleaned
- Water used was dust free
- Whenever a solution was changed, the tube was rinsed with the new solution under examination
- There was no air bubble inside the tube
- Reading was taken when halves of the field of view becomes equally illuminated.